Hierarchical Reinforcement Learning for Cryptocurrency Trading via Adaptive Strategy Selection

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Introduction

- Cryptocurrency markets operate continuously with deep liquidity in perpetual futures, making them ideal for algorithmic trading.
- Key challenges: non-stationarity, heavy-tailed returns, and frequent regime changes across macro and micro timescales.
- Fixed, single-strategy systems perform well in narrow conditions but fail when volatility, cross-asset relationships, or order-flow dynamics shift.
- Our approach: adaptive strategy selection where RL agents learn *when* each trading hypothesis is reliable, rather than learning policies end-to-end.
- We trade Solana (SOL) at two timescales: macro (daily) using spot and micro (sub-second) using perpetual futures.
- Cross-asset signals: BTC ($1.8T) and ETH ($0.4T) are far more liquid than SOL ($0.08T), creating exploitable lead-lag relationships.

Data and Feature Engineering

- All crypto data sourced from Binance Vision; S&P 500 from Yahoo Finance.
- **Macro Dataset**: 8-hour klines from Sept 2020 – Dec 2025 (1,899 samples). 16-dimensional feature vector: SOL momentum at 5 horizons, BTC/ETH/S&P momentum at 3 horizons, plus volume and price.
- **Micro Dataset**: 100ms aggregated trades for BTC, ETH, SOL perpetuals (Jan – Apr 2025). 420-dimensional state vector with returns, momentum (7 windows: 100ms–60min), flow imbalance, and realized volatility.
- Momentum defined as: $M_t = \frac{P_t - P_{t-\tau}}{P_{t-\tau}}$
- Lead-lag screening on held-out data confirmed ETH/BTC log returns correlate ∼0.12 with 100ms-ahead SOL returns.

Feature	Target	Corr.
eth_log_return	fwd_ret_100ms	0.120
btc_log_return	fwd_ret_100ms	0.114
eth_momentum_100ms	fwd_ret_100ms	0.111
sol_realized_vol_30s	fwd_vol_30s	0.525

Figure 1. Top feature correlations from lead-lag screening (targets are SOL).

Property	Macro / Micro
Total samples	1,899 / ∼10M
Features	17 / 420
Train/Val/Test	90/10 / 70/15/15
Action space	21 / 7+7

Figure 2. Dataset statistics for macro and micro trading systems.

Macro-Level Trading

- Trade SOL spot (not futures) to avoid liquidation risk over multi-day horizons.
- State: 17-dim vector = normalized inventory + 16 cross-asset momentum features.
- Action: Discrete inventory change $a_t \in \{-10, ..., +10\}$ with cash constraint.
- Reward: Log-transformed P&L to stabilize training:

$$r_t = \text{sign}(r_t^{\text{raw}}) \cdot \log(1 + |r_t^{\text{raw}}|), \quad r_t^{\text{raw}} = (p_{t+1} - p_t) \cdot I_{t+1}$$

- DQN with 2 hidden layers (128 units), experience replay (10K buffer), ϵ-greedy exploration with exponential decay.
- Q-learning update with target network (soft updates every 500 steps):

$$\mathcal{L}(\theta) = \mathbb{E}\left[\left(r_t + \gamma \max_{a'} Q_{\theta^-}(s_{t+1}, a') - Q_\theta(s_t, a_t)\right)^2\right]$$

Micro-Level Hierarchical Architecture

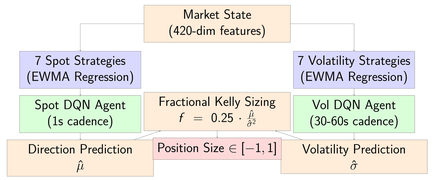

Figure 3. Hierarchical RL architecture with separate spot/volatility agents selecting from interpretable strategy families.

Spot Strategies (Direction)

- `flat`: Predict zero, abstain
- `btc_leads`: BTC → SOL
- `eth_leads`: ETH → SOL
- `both_lead`: Combined signal
- `sol_momentum`: Trend follows
- `sol_mean_revert`: Overreaction
- `flow_driven`: Order flow

Volatility Strategies (Risk)

- `flat`: Constant baseline
- `vol_30s_persists`: Clustering
- `vol_30s_mean_reverts`: Spikes fade
- `vol_1m_persists`: Longer horizon
- `vol_1m_mean_reverts`: Longer horizon
- `vol_cross_asset`: Contagion
- `vol_from_flow`: Flow → vol

- Key insight: Learn *when* each hypothesis applies rather than discovering price dynamics from scratch.
- Each strategy uses online EWMA regression with halflife $\tau = 600$s to adapt to non-stationarity:

$$\mathbf{\Sigma}_{XX}^{(t)} = \lambda \mathbf{\Sigma}_{XX}^{(t-1)} + (1-\lambda)\mathbf{x}_t\mathbf{x}_t^\top, \quad \mathbf{w} = (\mathbf{\Sigma}_{XX} + \alpha \mathbf{I})^{-1}\mathbf{\Sigma}_{Xy}$$

- Differential Sharpe Ratio (DSR) reward optimizes risk-adjusted returns directly:

$$D_t = \frac{B_{t-1}\Delta A_t - \frac{1}{2}A_{t-1}\Delta B_t}{(B_{t-1} - A_{t-1}^2)^{3/2} + \epsilon}$$

Training Results

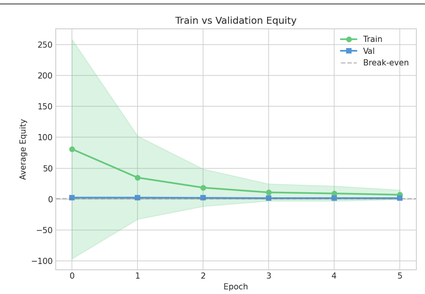

Figure 4. Train vs. validation equity curves over training epochs.

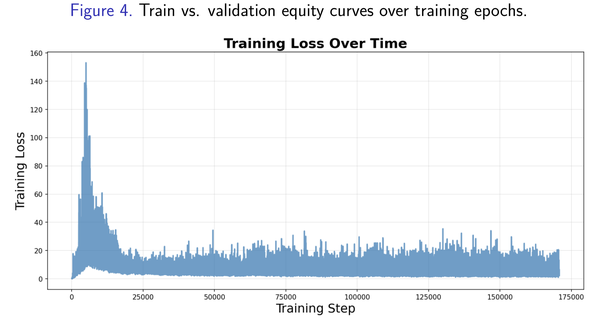

Figure 5. Macro agent training loss convergence.

Discussion and Analysis

- **Macro Results**: Agent achieved +6.11% return during a -21.5% bear market, outperforming buy-and-hold by 27.6%.
- **Micro Results**: Final equity 6.93×, Sharpe ≈ 0.98, with consistent small per-step edges compounding over many decisions.

Table 1. Macro agent evaluation (190 days)

Strategy	Final Value	Return
RL Agent	$10,611	+6.11%
Buy & Hold	$7,849	−21.51%
Bogo Trader	$9,815	−1.85%

Table 2. Micro agent test metrics

Metric	Value	Metric	Value
Final equity	6.93×	Sharpe	0.98
Mean step Δ	$+6.9\times10^{-5}$	Max DD	−1.64%
Max gain	+1.94%	Max loss	−1.15%

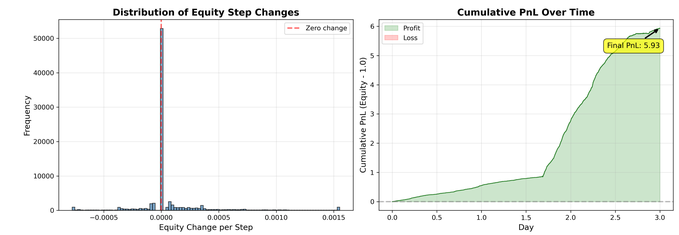

Figure 6. Test performance: step-level return distribution (left) and cumulative PnL (right).

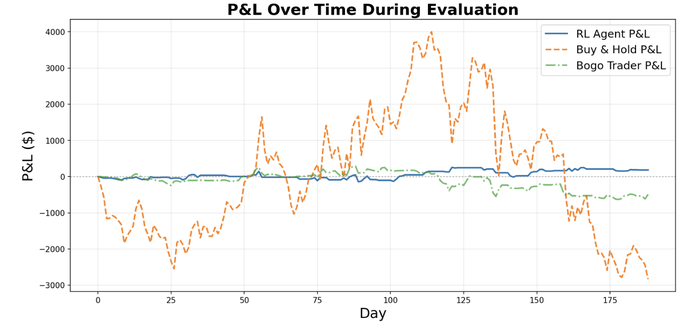

Figure 7. Macro agent P&L over 190-day evaluation period vs. baselines.

Conclusion and Future Work

- Demonstrated that hierarchical RL with interpretable strategy families can outperform static baselines in crypto markets.
- Key insight: Learning regime selection is more sample-efficient than end-to-end policy learning in low signal-to-noise environments.
- Cross-asset information (BTC/ETH) dominates short-horizon SOL direction prediction.
- Kelly sizing with volatility blending enables risk-aware position scaling.
- **Future directions**:
  - Transaction-cost-aware rewards and turnover penalties
  - Multi-asset portfolio generalization (joint BTC/ETH/SOL)
  - Position-aware state representations for inventory management
  - Extension to explicit volatility trading (options, straddles)

References

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